

Sketching

1 Mixed Poisson Distribution

In its simplest form, a Poisson distribution models the probability of a number of events occurring in a fixed interval, given that we know the average rate (λ) of those events. Its probability mass function is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

A mixed Poisson distribution occurs when the rate parameter itself, λ , is not a fixed number, but rather a random variable. This introduces an additional level of uncertainty.

- **Standard Poisson:** Imagine a set of very specific documents, such as only soccer game summaries. The word "goal" appears, on average, exactly 3 times per summary. The rate $\lambda = 3$ is fixed.
- **Mixed Poisson:** Now, imagine your dataset contains two types of documents: soccer game summaries (where the average rate of "goal" is $\lambda_{\text{soc}} = 3$) and finance articles (where the word "goal" is less common, perhaps $\lambda_{\text{fin}} = 0.5$). If you randomly pick a document without knowing its type, the rate λ is now a random variable (it could be 3 or 0.5, with some probability). The distribution of the word count for "goal" in this random document is a "mixture" of the two Poisson distributions.

The mixed Poisson distribution takes the following form:

$$X \mid Z = z \sim \bigotimes_{i=1}^d \text{Poisson}((\mathbf{A}\mathbf{z})_i)$$

- X is a vector of counts with d dimensions: $X = [X_1, X_2, \dots, X_d]^T$.
- Z is a vector of m latent (hidden) random variables that make up the "mixture".
- $\mathbf{A}$ is a weight matrix with d rows and m columns. Its elements are non-negative. Each row i of $\mathbf{A}$ defines how the latent factors in z combine to form the rate for the count X_i .
- $\mathbf{z}$ is a vector of m non-negative latent factors. This is the specific value assumed by the random variable Z .
- $(\mathbf{A}\mathbf{z})_i$ is the Poisson rate λ for the i -th count X_i . It is calculated by multiplying the i -th row of the matrix $\mathbf{A}$ by the vector $\mathbf{z}$. The complete vector of rates is $\lambda = \mathbf{A}\mathbf{z}$.
- $\bigotimes_{i=1}^d \text{Poisson}(\dots)$ represents the product of independent distributions. This means that, once we know $\mathbf{z}$, the d counts $(X_1, \dots, X_d)$ are all statistically independent of each other. Each X_i is drawn from its own Poisson distribution with its own rate $(\mathbf{A}\mathbf{z})_i$.

1.1 Example

Let's model the count of $d = 3$ words in documents, based on $m = 2$ latent topics.

- X_1 : count of "calculus"
- X_2 : count of "football"
- X_3 : count of "investment"
- z_1 : exposure to Topic 1 ("Academic")
- z_2 : exposure to Topic 2 ("Sports & Finance")
- z_3 : exposure to both topics

Let our weight matrix $\mathbf{A}$ be defined as:

$$\mathbf{A} = \begin{pmatrix} 10 & 1 \\ 2 & 8 \\ 1 & 9 \end{pmatrix}$$

Interpretation:

- Row 1 ("calculus"): $\begin{bmatrix} 10 & 1 \end{bmatrix}$ - Strongly associated with Topic 1.
- Row 2 ("football"): $\begin{bmatrix} 2 & 8 \end{bmatrix}$ - Strongly associated with Topic 2.
- Row 3 ("investment"): $\begin{bmatrix} 1 & 9 \end{bmatrix}$ - Strongly associated with Topic 2.

Let's generate samples of X for different documents (different z vectors).

Scenario 1: "Academic" Document

We assume a document with high exposure to Topic 1.

$$z_1 = \begin{pmatrix} 5 \\ 0.1 \end{pmatrix}$$

We calculate the rates $\lambda = \mathbf{A}z_1$:

$$\lambda_1 = \begin{pmatrix} 10 & 1 \\ 2 & 8 \\ 1 & 9 \end{pmatrix} \begin{pmatrix} 5 \\ 0.1 \end{pmatrix} = \begin{pmatrix} 50.1 \\ 10.8 \\ 5.9 \end{pmatrix}$$

The samples of X (word counts) will come from $X_1 \sim \text{Poisson}(50.1)$, $X_2 \sim \text{Poisson}(10.8)$, and $X_3 \sim \text{Poisson}(5.9)$. The "calculus" counts are high, as expected.

Scenario 2: "Sports & Finance" Document

We assume a document with high exposure to Topic 2.

$$z_2 = \begin{pmatrix} 0.5 \\ 10 \end{pmatrix}$$

We calculate the rates $\lambda = \mathbf{A}z_2$:

$$\lambda_2 = \begin{pmatrix} 10 & 1 \\ 2 & 8 \\ 1 & 9 \end{pmatrix} \begin{pmatrix} 0.5 \\ 10 \end{pmatrix} = \begin{pmatrix} 15 \\ 81 \\ 90.5 \end{pmatrix}$$

The samples of X will come from $X_1 \sim \text{Poisson}(15)$, $X_2 \sim \text{Poisson}(81)$, and $X_3 \sim \text{Poisson}(90.5)$. The "football" and "investment" counts are high.

Scenario 3: Document with a Balanced Mixture

We assume a document with moderate exposure to both topics.

$$z_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

We calculate the rates $\lambda = \mathbf{A}z_3$:

$$\lambda_3 = \begin{pmatrix} 10 & 1 \\ 2 & 8 \\ 1 & 9 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 34 \\ 38 \\ 39 \end{pmatrix}$$

The samples of X will come from $X_1 \sim \text{Poisson}(34)$, $X_2 \sim \text{Poisson}(38)$, and $X_3 \sim \text{Poisson}(39)$. All counts are similar.

2 Probability-Generating Function (PGF)

For a d -dimensional count vector $\mathbf{X} = (X_1, \dots, X_d)$, the PGF $G_{\mathbf{X}}(\mathbf{t})$ is defined as the expectation of a product of powers of the components of $\mathbf{t} = (t_1, \dots, t_d)$:

$$G_{\mathbf{X}}(\mathbf{t}) = \mathbb{E} \left[\prod_{i=1}^d t_i^{X_i} \right] = \mathbb{E} \left[e^{\langle \mathbf{X}, \ln(\mathbf{t}) \rangle} \right]$$

Considering the expectancy estimation $\hat{\mathbb{E}}[\mathbf{X}] = \frac{1}{n} \sum_{j=1}^n \mathbf{X}_j$, where j are the samples of $\mathbf{X}$, it is possible to estimate the PGF from a dataset:

$$\hat{G}_{\mathbf{X}}(\mathbf{t}) = \frac{1}{n} \sum_{j=1}^n e^{\langle \mathbf{X}^{(j)}, \ln(\mathbf{t}) \rangle}$$

Poisson PGF (Univariate)

For a single univariate Poisson random variable X with a fixed mean (rate) μ , the PGF is defined as $G_X(t) = \mathbb{E}[t^X]$. Using the Poisson probability mass function $P(X = n) = e^{-\mu} \frac{\mu^n}{n!}$, the closed form is:

$$\begin{aligned} G_X(t) &= \sum_{n=0}^{\infty} P(X = n) t^n = \sum_{n=0}^{\infty} e^{-\mu} \frac{\mu^n}{n!} t^n \\ &= e^{-\mu} \sum_{n=0}^{\infty} \frac{(\mu t)^n}{n!} \\ &= e^{-\mu} e^{\mu t} = e^{\mu(t-1)} \end{aligned}$$

Recognizing the Taylor series expansion for the exponential function, $e^a = \sum_{n=0}^{\infty} \frac{a^n}{n!}$ where $a = \mu t$, the PGF simplifies to:

$$G_X(t) = e^{\mu(t-1)}$$

Mixed Poisson PGF (Multivariate)

Let's consider a multivariate observation $\mathbf{X}$ generated by a mixture model, where the conditional distribution of $\mathbf{X}$ given a latent random vector $\mathbf{Z} = \mathbf{z}$ (linked to the rates) is:

$$\mathbf{X} | \mathbf{Z} = \mathbf{z} \sim \bigotimes_{i=1}^d \text{Poisson}((\mathbf{A}\mathbf{z})_i)$$

The conditional PGF for the multivariate $\mathbf{X}$ is the product of the independent marginal PGFs, where $\mathbf{A}\mathbf{z}$ is the vector of Poisson rates:

$$G_{\mathbf{X}|\mathbf{Z}=\mathbf{z}}(\mathbf{t}) = \mathbb{E} \left[\prod_{i=1}^d t_i^{X_i} \mid \mathbf{Z} = \mathbf{z} \right] = \prod_{i=1}^d e^{(\mathbf{A}\mathbf{z})_i (t_i - 1)}$$

This simplifies using vector notation $\langle \cdot, \cdot \rangle$ (and $\mathbf{1}$ as a vector of ones):

$$G_{\mathbf{X}|\mathbf{Z}=\mathbf{z}}(\mathbf{t}) = e^{\langle \mathbf{A}\mathbf{z}, \mathbf{t} - \mathbf{1} \rangle}$$

The mixed Poisson PGF is obtained by taking the expectation of the conditional PGF $G_{\mathbf{X}|\mathbf{Z}=\mathbf{z}}(\mathbf{t})$ over the distribution of the latent mixing variable $\mathbf{Z}$.

Assuming the discrete mixture model defined in the text, where $\mathbf{Z}$ takes one of K fixed values $\mathbf{z}_k$ with probability π_k :

$$p(\mathbf{z}) = \sum_{k=1}^K \pi_k \delta_{\mathbf{z}_k}(\mathbf{z})$$

Then,

$$G_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}_{\mathbf{Z}}[G_{\mathbf{X}|\mathbf{Z}=\mathbf{z}}(\mathbf{t})] = \sum_{k=1}^K \pi_k G_{\mathbf{X}|\mathbf{Z}=\mathbf{z}_k}(\mathbf{t})$$

Substituting the conditional PGF yields the closed-form expression for the Mixed Poisson PGF:

$$G_{\mathbf{X}}(\mathbf{t}) = \sum_{k=1}^K \pi_k e^{\langle \mathbf{A}\mathbf{z}_k, \mathbf{t} - \mathbf{1} \rangle} = \sum_{k=1}^K \pi_k e^{\langle \boldsymbol{\lambda}_k, \mathbf{t} - \mathbf{1} \rangle}$$

where $\boldsymbol{\lambda}_k = \mathbf{A}\mathbf{z}_k$ is the vector of Poisson rates for the k -th mixture component.

Sampling Strategy

We sample the Probability-Generating Function (PGF) along complex lines defined by a direction vector $\mathbf{u}$, a base point $\mathbf{p}$, and a step index n . We scale the sampling by a factor Δ that is around $1/\lambda_{max}$ (highest poisson rate give by $\mathbf{Az}$). We define the sequence of sampling points $\mathbf{t}(\mathbf{p}, \mathbf{u}, n) \in \mathbb{C}^d$ as:

$$\mathbf{t}(\mathbf{p}, \mathbf{u}, n) = \mathbf{1}_d + j\Delta(\mathbf{p} + n\mathbf{u}), \quad n = 0, \dots, N-1$$

Substituting this into the PGF definition yields the sequence of samples $y[n]$:

$$\begin{aligned} y[n] = G_{\mathbf{X}}(\mathbf{t}(\mathbf{p}, \mathbf{u}, n)) &= \sum_{k=1}^K \pi_k e^{\langle \boldsymbol{\lambda}_k, (\mathbf{1}_d + j\Delta(\mathbf{p} + n\mathbf{u})) - \mathbf{1}_d \rangle} \\ &= \sum_{k=1}^K \pi_k e^{j\langle \Delta\boldsymbol{\lambda}_k, \mathbf{p} + n\mathbf{u} \rangle} \end{aligned}$$

This equation maps exactly to a multi-dimensional harmonic retrieval signal model evaluated along a line coordinate $\mathbf{n} = \mathbf{p} + n\mathbf{u}$:

- The mixture probabilities π_k correspond to the amplitudes a_k .
- The scaled rates $\Delta\boldsymbol{\lambda}_k$ correspond to the spatial frequencies $\boldsymbol{\omega}_k$.
- The sequence of samples $y[n]$ corresponds to the signal observations along the line.

Thus, recovering the frequencies $\{\Delta\boldsymbol{\lambda}_k\}$ from the observed samples is precisely the spectral estimation problem that the JOINT ESPRIT algorithm is designed to solve.

3 The ESPRIT algorithm

It is logical to first define the one dimensional ESPRIT problem before tackling the multidimensional Joint ESPRIT algorithm.

The Signal Model and Array Manifold

We begin by defining the data received by an array of M sensors. Let's assume D narrowband signals, $s_1(t), \dots, s_D(t)$, impinge on a uniform linear array (ULA) from directions $\theta_1, \dots, \theta_D$. The signal received at the m -th sensor is a superposition of all D signals plus additive noise $n_m(t)$.

The full $M \times 1$ received signal vector $\mathbf{x}(t)$ is given by:

$$\mathbf{x}(t) = \sum_{k=1}^D \mathbf{a}(\theta_k) s_k(t) + \mathbf{n}(t) = \mathbf{A} \mathbf{s}(t) + \mathbf{n}(t)$$

Where:

- $\mathbf{x}(t) = [x_1(t), \dots, x_M(t)]^T$ is the $M \times 1$ signal vector measured by the M different sensors.
- $\mathbf{s}(t) = [s_1(t), \dots, s_D(t)]^T$ is the $D \times 1$ vector of source signals.
- $\mathbf{n}(t) = [n_1(t), \dots, n_M(t)]^T$ is the $M \times 1$ noise vector.
- $\mathbf{A} = [\mathbf{a}(\theta_1) \cdots \mathbf{a}(\theta_D)]$ is the $M \times D$ array manifold matrix.

For a ULA with inter-element spacing d and signal wavelength λ , the steering vector $\mathbf{a}(\theta_k)$ for the k -th signal is:

$$\mathbf{a}(\theta_k) = \begin{bmatrix} 1 \\ e^{j\phi_k} \\ e^{j2\phi_k} \\ \vdots \\ e^{j(M-1)\phi_k} \end{bmatrix}, \text{ where } \phi_k = \frac{2\pi}{\lambda} d \sin(\theta_k)$$

The full manifold matrix $\mathbf{A}$ is formed by concatenating these steering vectors as columns: $\mathbf{A} = [\mathbf{a}(\theta_1) \cdots \mathbf{a}(\theta_D)]$. Explicitly, this $M \times D$ matrix has a Vandermonde structure:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e^{j\phi_1} & e^{j\phi_2} & \cdots & e^{j\phi_D} \\ e^{j2\phi_1} & e^{j2\phi_2} & \cdots & e^{j2\phi_D} \\ \vdots & \vdots & \ddots & \vdots \\ e^{j(M-1)\phi_1} & e^{j(M-1)\phi_2} & \cdots & e^{j(M-1)\phi_D} \end{bmatrix}$$

The Core Principle: Rotational Invariance

The ESPRIT algorithm's ingenuity lies in its use of a specific array geometry. The array is partitioned into two identical, overlapping subarrays, which we will call X and Y .

- **Subarray X:** Consists of sensors 1, 2, $\dots$, $(M-1)$.
- **Subarray Y:** Consists of sensors 2, 3, $\dots$, M .

Subarray Y is a perfect copy of Subarray X , but translationally displaced by the distance d . This physical displacement creates a mathematical relationship between their respective manifold matrices, $\mathbf{A}_X$ and $\mathbf{A}_Y$.

$\mathbf{A}_X$ consists of the first $M-1$ rows of $\mathbf{A}$, and $\mathbf{A}_Y$ consists of the last $M-1$ rows of $\mathbf{A}$. Let's look at the k -th column of $\mathbf{A}_Y$ (the steering vector for θ_k on subarray Y):

$$\mathbf{a}_Y(\theta_k) = \begin{bmatrix} e^{j\phi_k} \\ e^{j2\phi_k} \\ \vdots \\ e^{j(M-1)\phi_k} \end{bmatrix} = e^{j\phi_k} \begin{bmatrix} 1 \\ e^{j\phi_k} \\ \vdots \\ e^{j(M-2)\phi_k} \end{bmatrix} = (e^{j\frac{2\pi}{\lambda} d \sin(\theta_k)}) \cdot \mathbf{a}_X(\theta_k)$$

This shows that the steering vector for Subarray Y is just the steering vector for Subarray X multiplied by a complex phase factor $e^{j\phi_k}$.

We can express this relationship for all D signals simultaneously using a $D \times D$ diagonal matrix Φ :

$$\mathbf{A}_Y = \mathbf{A}_X \Phi \tag{1}$$

where

$$\Phi = \text{diag}(e^{j\phi_1}, e^{j\phi_2}, \dots, e^{j\phi_D})$$

This is the rotational invariance property. The DOAs we seek are embedded in the diagonal elements of Φ .

Subspace Decomposition and Equivalence

The next step is to estimate the signal subspace from the received data. We compute the $M \times M$ covariance matrix $\mathbf{R}_{xx}$, theoretically defined as:

$$\mathbf{R}_{xx} = E [\mathbf{x}(t)\mathbf{x}(t)^H]$$

In practice, we estimate this matrix from n discrete time samples (snapshots). We form a data matrix $\mathbf{X}$ of size $n \times M$, where each of the n rows contains the M sensor measurements for a given snapshot. The sample covariance matrix $\hat{\mathbf{R}}_{xx}$ is then computed as:

$$\hat{\mathbf{R}}_{xx} = \frac{1}{n} \mathbf{X}^T \mathbf{X}^*$$

Assuming the signals and noise are uncorrelated ($E[\mathbf{s}\mathbf{n}^H] = \mathbf{0}$) and the noise is spatially white ($E[\mathbf{n}\mathbf{n}^H] = \sigma^2 \mathbf{I}$), the theoretical covariance matrix simplifies to:

$$\mathbf{R}_{xx} = E [(\mathbf{A}\mathbf{s}(t) + \mathbf{n}(t))(\mathbf{A}\mathbf{s}(t) + \mathbf{n}(t))^H] = \mathbf{A}E[\mathbf{s}(t)\mathbf{s}(t)^H]\mathbf{A}^H + \sigma^2 \mathbf{I} = \mathbf{A}\mathbf{R}_{ss}\mathbf{A}^H + \sigma^2 \mathbf{I}$$

We perform an eigendecomposition on our estimated matrix $\mathbf{R}_{xx}$. The matrix $\mathbf{R}_{xx}$ has M eigenvalues.

- D eigenvalues will be $\lambda_k + \sigma^2$, where λ_k are the eigenvalues of $\mathbf{A}\mathbf{R}_{ss}\mathbf{A}^H$.
- $M - D$ eigenvalues will be equal to σ^2 .

The D eigenvectors corresponding to the largest eigenvalues form the basis for the **Signal Subspace**, $\mathbf{E}_S$. The other $M - D$ eigenvectors form the **Noise Subspace**, $\mathbf{E}_N$.

Crucially, the eigenvectors of $\mathbf{R}_{xx}$ are the same as the eigenvectors of $\mathbf{A}\mathbf{R}_{ss}\mathbf{A}^H$. The column space of $\mathbf{A}\mathbf{R}_{ss}\mathbf{A}^H$ is identical to the column space of $\mathbf{A}$ (assuming D non-correlated signals). Therefore, the signal subspace and the array manifold span the **same D -dimensional space**.

$$\text{span}(\mathbf{E}_S) = \text{span}(\mathbf{A})$$

This means there exists a unique, $D \times D$ invertible matrix $\mathbf{T}$ that relates them:

$$\mathbf{E}_S = \mathbf{A}\mathbf{T} \quad (2)$$

This $\mathbf{T}$ matrix is unknown, but as we will see, we don't need to find it.

Deriving the Solution

Now we tie everything together. We partition our estimated signal subspace $\mathbf{E}_S$ into two $(M - 1) \times D$ matrices, just as we did with the array manifold:

- $\mathbf{E}_X$: First $M - 1$ rows of $\mathbf{E}_S$.
- $\mathbf{E}_Y$: Last $M - 1$ rows of $\mathbf{E}_S$.

Let's substitute $\mathbf{E}_S = \mathbf{A}\mathbf{T}$ (2) into these new matrices.

$$\mathbf{E}_X = \mathbf{J}_X \mathbf{E}_S = (\mathbf{J}_X \mathbf{A})\mathbf{T} = \mathbf{A}_X \mathbf{T} \quad (3)$$

$$\mathbf{E}_Y = \mathbf{J}_Y \mathbf{E}_S = (\mathbf{J}_Y \mathbf{A})\mathbf{T} = \mathbf{A}_Y \mathbf{T} \quad (4)$$

(where $\mathbf{J}_X$ and $\mathbf{J}_Y$ are the selection matrices of 1s and 0s that select the rows).

Now, we use the rotational invariance property $\mathbf{A}_Y = \mathbf{A}_X \Phi$ (1) by substituting it into (4):

$$\mathbf{E}_Y = (\mathbf{A}_X \Phi)\mathbf{T} = \mathbf{A}_X \Phi \mathbf{T} \quad (5)$$

From (3), we can write $\mathbf{A}_X = \mathbf{E}_X \mathbf{T}^{-1}$ (since $\mathbf{E}_X$ and $\mathbf{T}$ are invertible). Let's substitute this into (5):

$$\mathbf{E}_Y = (\mathbf{E}_X \mathbf{T}^{-1}) \Phi \mathbf{T}$$

By re-grouping the matrices, we get the central ESPRIT equation:

$$\mathbf{E}_Y = \mathbf{E}_X (\mathbf{T}^{-1} \Phi \mathbf{T})$$

Let us define a new $D \times D$ matrix $\Psi = \mathbf{T}^{-1} \Phi \mathbf{T}$. Our equation becomes:

$$\mathbf{E}_Y = \mathbf{E}_X \Psi$$

Since $\mathbf{E}_X$ and $\mathbf{E}_Y$ are known (estimated from our data), we can solve this overdetermined system for Ψ . This is typically done using a Total Least Squares (TLS) solution, which is robust to noise.

$$\Psi_{TLS} = \text{argmin}_{\Psi} \|[\mathbf{E}_X \mid \mathbf{E}_Y]\|_F$$

Finding the DOAs from Eigenvalues

The final step is to retrieve the DOAs. We have our computed matrix $\mathbf{\Psi}$. The equation $\mathbf{\Psi} = \mathbf{T}^{-1}\mathbf{\Phi}\mathbf{T}$ is a **similarity transformation**. A fundamental theorem of linear algebra states that similar matrices have the exact same eigenvalues.

Therefore:

$$\text{eigenvalues}(\mathbf{\Psi}) = \text{eigenvalues}(\mathbf{\Phi}) = \{e^{j\phi_1}, e^{j\phi_2}, \dots, e^{j\phi_D}\}$$

So, by computing the D eigenvalues of our estimated matrix $\mathbf{\Psi}$, $\{\lambda_1, \dots, \lambda_D\}$, we have found the D phase factors:

$$\lambda_k = e^{j\phi_k} = e^{j\frac{2\pi}{\lambda}d\sin(\theta_k)}$$

We can now find each DOA θ_k by taking the angle of the corresponding eigenvalue:

$$\arg(\lambda_k) = \phi_k = \frac{2\pi}{\lambda}d\sin(\theta_k)$$

$$\theta_k = \arcsin\left(\frac{\lambda \cdot \arg(\lambda_k)}{2\pi d}\right)$$

This gives us the D direction of arrivals without any computationally expensive spectral search, which is the main advantage of ESPRIT.

4 The Joint ESPRIT Algorithm

Signal Model

We consider a d -dimensional signal $\mathbf{y}[\mathbf{n}]$ composed of r complex components. The signal at any d -dimensional coordinate $\mathbf{n}$ is sampled with a grid scale Δ :

$$\mathbf{y}[\mathbf{n}] = \sum_{k=1}^r a_k e^{j\langle \Delta \boldsymbol{\omega}_k, \mathbf{n} \rangle} + \mathbf{w}[\mathbf{n}]$$

where:

- $a_k \in \mathbb{C}$ is the complex amplitude of component k .
- $\boldsymbol{\omega}_k$ is the unknown d -dimensional frequency vector we want to find.
- Δ is a known grid scale factor.
- $\mathbf{w}[\mathbf{n}]$ is additive noise.

Acquisition Design

Instead of sampling the entire d -Dimensional space, we sample it along a set of M 1D lines, where $M \geq d$.

1. Choose a set of M direction vectors, $\mathcal{U} = \{\mathbf{u}_l\}_{l=1}^M$, where $\mathbf{u}_l \in \mathbb{Z}^d$.
2. For each direction $\mathbf{u}_l$, choose S_l random "base points" $\{\mathbf{p}_{l,s}\}_{s=1}^{S_l}$, with $S_l \geq r$.
3. For each pair (l, s) , sample N_l points along the line $\mathcal{L}_{l,s}$, with $N_l \geq r + 1$.

$$\mathcal{L}_{l,s} = \{\mathbf{p}_{l,s} + n\mathbf{u}_l : n = 0, \dots, N_l - 1\}$$

This strategy reduces the d -D problem to a set of M 1D problems, one for each direction $\mathbf{u}_l$.

4.1 Step 1: Per-Direction Model (Multi-snapshot 1D)

For a single direction l , we analyze the signal $z_{l,s}[n]$ sampled along the line (l, s) . By substituting the line coordinate $\mathbf{n}' = \mathbf{p}_{l,s} + n\mathbf{u}_l$ into the signal model:

$$\begin{aligned} z_{l,s}[n] &= \sum_{k=1}^r a_k e^{j\langle \Delta \boldsymbol{\omega}_k, \mathbf{p}_{l,s} + n\mathbf{u}_l \rangle} + w_{l,s}[n] \\ z_{l,s}[n] &= \sum_{k=1}^r \left(a_k e^{j\langle \Delta \boldsymbol{\omega}_k, \mathbf{p}_{l,s} \rangle} \right) \left(e^{j\varphi_{k,l}} \right)^n + w_{l,s}[n] \end{aligned}$$

This is a standard 1D multi-snapshot model. The term $\varphi_{k,l} = \Delta \langle \boldsymbol{\omega}_k, \mathbf{u}_l \rangle$ is the 1D frequency for this direction; it is the scaled scalar projection of the d -D frequency vector $\boldsymbol{\omega}_k$ onto the 1D direction vector $\mathbf{u}_l$. We can stack all S_l snapshots (from the S_l base points) into a matrix equation:

$$\mathbf{Z}_l = \mathbf{V}_l \mathbf{C}_l + \mathbf{W}_l$$

where $\mathbf{Z}_l$ is the $N_l \times S_l$ measurement matrix, $\mathbf{V}_l$ is the $N_l \times r$ frequency matrix, $\mathbf{C}_l$ is the $r \times S_l$ amplitude matrix, and $\mathbf{W}_l$ is the noise matrix. Explicitly, the matrices are:

$$\mathbf{Z}_l = \begin{pmatrix} z_{l,1}[0] & z_{l,2}[0] & \cdots & z_{l,S_l}[0] \\ z_{l,1}[1] & z_{l,2}[1] & \cdots & z_{l,S_l}[1] \\ \vdots & \vdots & \ddots & \vdots \\ z_{l,1}[N_l-1] & z_{l,2}[N_l-1] & \cdots & z_{l,S_l}[N_l-1] \end{pmatrix}$$

$$\mathbf{V}_l = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ e^{j\varphi_{1,l}} & e^{j\varphi_{2,l}} & \cdots & e^{j\varphi_{r,l}} \\ e^{j2\varphi_{1,l}} & e^{j2\varphi_{2,l}} & \cdots & e^{j2\varphi_{r,l}} \\ \vdots & \vdots & \ddots & \vdots \\ e^{j(N_l-1)\varphi_{1,l}} & e^{j(N_l-1)\varphi_{2,l}} & \cdots & e^{j(N_l-1)\varphi_{r,l}} \end{pmatrix} \quad \text{where } \varphi_{k,l} = \Delta \langle \boldsymbol{\omega}_k, \mathbf{u}_l \rangle$$

$$\mathbf{C}_l = \begin{pmatrix} a_1 e^{j\langle \Delta \boldsymbol{\omega}_1, \mathbf{p}_{1,l} \rangle} & a_1 e^{j\langle \Delta \boldsymbol{\omega}_1, \mathbf{p}_{2,l} \rangle} & \cdots & a_1 e^{j\langle \Delta \boldsymbol{\omega}_1, \mathbf{p}_{S_l,l} \rangle} \\ a_2 e^{j\langle \Delta \boldsymbol{\omega}_2, \mathbf{p}_{1,l} \rangle} & a_2 e^{j\langle \Delta \boldsymbol{\omega}_2, \mathbf{p}_{2,l} \rangle} & \cdots & a_2 e^{j\langle \Delta \boldsymbol{\omega}_2, \mathbf{p}_{S_l,l} \rangle} \\ \vdots & \vdots & \ddots & \vdots \\ a_r e^{j\langle \Delta \boldsymbol{\omega}_r, \mathbf{p}_{1,l} \rangle} & a_r e^{j\langle \Delta \boldsymbol{\omega}_r, \mathbf{p}_{2,l} \rangle} & \cdots & a_r e^{j\langle \Delta \boldsymbol{\omega}_r, \mathbf{p}_{S_l,l} \rangle} \end{pmatrix}$$

4.2 Step 2: Subspace Estimation

For each direction l , we compute the spatial covariance matrix and find the signal subspace.

1. Compute covariance: $\hat{\mathbf{R}}_l = \frac{1}{S_l} \mathbf{Z}_l \mathbf{Z}_l^H$.
2. Compute the SVD of $\hat{\mathbf{R}}_l$ and extract the r dominant eigenvectors to form the $N_l \times r$ signal subspace matrix, $\mathbf{U}_{s,l}$.

4.3 Step 3: 1D ESPRIT per Direction

We apply the core ESPRIT rotational invariance principle to each $\mathbf{U}_{s,l}$ independently.

1. Partition $\mathbf{U}_{s,l}$ into two $(N_l - 1) \times r$ matrices:
 - $\mathbf{U}_1 = \mathbf{U}_{s,l}$ with the last row removed.
 - $\mathbf{U}_2 = \mathbf{U}_{s,l}$ with the first row removed.
2. Solve the linear system $\mathbf{U}_2 \approx \mathbf{U}_1 \boldsymbol{\Psi}_l$ (e.g., via least-squares) to find the $r \times r$ "shuffled" matrix $\boldsymbol{\Psi}_l$.

At the end of this step, we have M calculated, "shuffled" matrices: $\{\boldsymbol{\Psi}_1, \boldsymbol{\Psi}_2, \dots, \boldsymbol{\Psi}_M\}$. This is the dead end of the naive approach. If we call a standard eigenvalue solver on each $\boldsymbol{\Psi}_l$, it will return r eigenvalues. Crucially, it returns them in a numerical order (e.g., sorted by magnitude), not in their true physical order (index $k = 1, \dots, r$). Because the eigenvalues for each $\boldsymbol{\Psi}_l$ have different magnitudes, this numerical sorting scrambles the component order differently for each direction, and we are left with the "pairing problem".

4.4 Step 4: Joint ESPRIT (Simultaneous Diagonalization)

From the 1D ESPRIT case, we know the relationship between the calculated matrix $\boldsymbol{\Psi}_l$ and its corresponding diagonal matrix of phase factors $\boldsymbol{\Phi}_l$ is a similarity transformation $\boldsymbol{\Psi}_l = \mathbf{T}^{-1} \boldsymbol{\Phi}_l \mathbf{T}$.

The core theory of Joint ESPRIT states that, in the absence of noise, this similarity transform $\mathbf{T}$ is common to all M directions. Therefore, all M matrices $\{\boldsymbol{\Psi}_1, \dots, \boldsymbol{\Psi}_M\}$ share a common set of eigenvectors (the columns of $\mathbf{T}^{-1}$).

The "pairing problem" noted in Step 3 arises from this observation: if one applies a standard eigenvalue solver independently to each $\boldsymbol{\Psi}_l$, the solver will find the correct eigenvalues but return them in a numerical order (e.g., sorted by magnitude). This independent sorting breaks the common physical ordering of the components, making it impossible to pair the 1D frequencies across different directions.

Joint ESPRIT avoids this ambiguity by not finding the eigenvalues of each $\boldsymbol{\Psi}_l$ separately. Instead, it takes the entire set of "shuffled" matrices $\{\boldsymbol{\Psi}_1, \dots, \boldsymbol{\Psi}_M\}$ and searches for the single transformation matrix $\hat{\mathbf{T}}$ that best diagonalizes all of them simultaneously.

We solve the optimization problem:

$$\hat{\mathbf{T}} = \arg \min_{\mathbf{T}} \sum_{l=1}^M \left\| \text{offdiag}(\hat{\mathbf{T}} \boldsymbol{\Psi}_l \hat{\mathbf{T}}^{-1}) \right\|_F^2$$

This optimization finds the single $\hat{\mathbf{T}}$ that minimizes the off-diagonal elements across all matrices. This process is a form of simultaneous eigendecomposition and is formally known as **Joint Diagonalization**.

This algorithm finds the single "best-fit" un-shuffling matrix $\hat{\mathbf{T}}$ that is common to all M directions. This $\hat{\mathbf{T}}$ is our estimate for the "true" transformation matrix, and its inverse, $\hat{\mathbf{T}}^{-1}$, contains the correctly-ordered, common eigenvectors as its columns.

4.5 Step 5: Reconstruct d -D Frequencies

Now that we have the common un-shuffler $\hat{\mathbf{T}}$, we can find the paired frequencies.

1. **Un-shuffle the matrices:** We apply the common un-shuffler $\hat{\mathbf{T}}$ to all M of our calculated Ψ_l matrices to get the diagonalized "goal" matrices, $\hat{\Phi}_l$.

$$\hat{\Phi}_l = \hat{\mathbf{T}}\Psi_l\hat{\mathbf{T}}^{-1}$$

2. **Read the paired frequencies:** Because we used the single, common un-shuffler $\hat{\mathbf{T}}$, the diagonal elements of all the $\hat{\Phi}_l$ matrices are now correctly paired. The k -th diagonal element of $\hat{\Phi}_1$ corresponds to the k -th diagonal element of $\hat{\Phi}_2$, and so on. We find the 1D frequencies $\hat{\varphi}_{k,l}$ by taking the angle of the k -th diagonal element of $\hat{\Phi}_l$:

$$\hat{\varphi}_{k,l} = \arg\left((\hat{\Phi}_l)_{kk}\right)$$

3. **Phase Unwrapping:** It is necessary to "unwrap" the $\hat{\varphi}_{k,l}$ values since they were recovered using the $\arg()$ function, which only returns a "wrapped" value in the principal range (e.g., $(-\pi, \pi]$). Phase unwrapping algorithms correct these jumps by adding or subtracting multiples of 2π to restore a smooth sequence, $\tilde{\varphi}_k$. This process relies on a key assumption: the true frequency projections $\Delta\langle\omega_k, \mathbf{u}_l\rangle$ change smoothly (by less than π) between adjacent direction vectors. This assumption is only met if the set of M direction vectors is sufficiently dense, which is why M is chosen to be greater than d ($M > d$). This redundancy provides a dense sampling of the direction space, ensuring the projections can be unwrapped reliably.
4. **Reconstruct ω_k :** For each component k , we now have its vector of M unwrapped 1D frequency projections, $\tilde{\varphi}_k = [\tilde{\varphi}_{k,1}, \dots, \tilde{\varphi}_{k,M}]^T$. We find the d -D vector ω_k by solving the linear system $\tilde{\varphi}_k \approx \Delta\mathbf{U}\omega_k$. Explicitly, this is:

$$\underbrace{\begin{pmatrix} \tilde{\varphi}_{k,1} \\ \tilde{\varphi}_{k,2} \\ \vdots \\ \tilde{\varphi}_{k,M} \end{pmatrix}}_{\tilde{\varphi}_k \text{ (known)}} \approx \Delta \underbrace{\begin{pmatrix} -\mathbf{u}_1- \\ -\mathbf{u}_2- \\ \vdots \\ -\mathbf{u}_M- \end{pmatrix}}_{\mathbf{U} \text{ (known)}} \underbrace{\begin{pmatrix} \omega_{k,1} \\ \vdots \\ \omega_{k,d} \end{pmatrix}}_{\omega_k \text{ (unknown)}}$$

We solve this $M \times d$ system using the unwrapped vector $\tilde{\varphi}_k$:

$$\hat{\omega}_k = \frac{1}{\Delta}(\mathbf{U}^T\mathbf{U})^{-1}\mathbf{U}^T\tilde{\varphi}_k$$

4.6 Step 6: Amplitude Estimation

Now that we have the d -D frequencies $\hat{\omega}_k$ for all components, the only unknowns are the original amplitudes a_k . We solve for them by setting up one final global least-squares problem:

$$\hat{\mathbf{a}} = \arg \min_{\mathbf{a}} \sum_{\mathbf{n} \in \text{all sampled points}} \left| z[\mathbf{n}] - \sum_{k=1}^r a_k e^{j\langle \Delta\hat{\omega}_k, \mathbf{n} \rangle} \right|^2$$

This finds the r amplitudes that best fit all the data we collected.

4.7 Implementation adjustments

Although the standard derivation in Section 4 suggests estimating subspaces independently for each direction, this method fails to preserve the coherent basis structure required for joint diagonalization. Consequently, we replace the independent estimation with a Global SVD approach (detailed below). Additionally, we omit the phase unwrapping step described in the theory, as we found it degraded performance.

In the standard element-wise ESPRIT approach, one would compute the Singular Value Decomposition (SVD) for each measurement matrix Z_l individually to find a signal subspace U_l . However, the signal subspace is only unique up to a unitary rotation. If U_l is a valid basis for the signal subspace of direction l , then $U_l Q$ is also a valid basis for any unitary matrix Q .

When these subspaces are estimated independently, each direction l effectively chooses a different, arbitrary rotation Q_l . Consequently, the Rotational Invariance Matrices (Ψ_l) derived from these subspaces are expressed in different coordinate systems. This makes it impossible to find a single common diagonalizing matrix T in the Joint ESPRIT step, as the eigenvectors of Ψ_l and Ψ_k are unrelated.

To solve this, we enforce a single common coordinate system by estimating a global signal subspace from all data simultaneously.

1. Construction of Global Data Matrix

We interpret the data from all M directions as being generated by a single large "virtual array" observing the same S source snapshots. We vertically stack the M measurement matrices $Z_l \in \mathbb{C}^{N \times S}$ into a single global data matrix $X_{\text{glob}} \in \mathbb{C}^{MN \times S}$:

$$X_{\text{glob}} = \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_M \end{bmatrix}$$

2. Global Subspace Extraction

We perform a single Truncated SVD on this global matrix:

$$X_{\text{glob}} \approx U_{\text{glob}} \Sigma_{\text{glob}} V_{\text{glob}}^H$$

where $U_{\text{glob}} \in \mathbb{C}^{MN \times r}$ contains the r dominant left singular vectors. The columns of U_{glob} span the common signal subspace shared by all directions. Crucially, this fixes the basis rotation for the entire dataset.

3. Extraction of Directional Subspaces

The global subspace matrix U_{glob} is composed of M stacked blocks, each of size $N \times r$. We partition it back into blocks corresponding to each direction:

$$U_{\text{glob}} = \begin{bmatrix} \hat{U}_1 \\ \hat{U}_2 \\ \vdots \\ \hat{U}_M \end{bmatrix} \quad \text{where } \hat{U}_l \in \mathbb{C}^{N \times r}$$

Here, $\hat{U}_l$ represents the signal subspace for direction l , but unlike the independently estimated U_l , these $\hat{U}_l$ blocks are locked to the same global coordinates. They are coherent projections of the single global signal structure.

4. Computation of Coherent RIMs

For each block $\hat{U}_l$, we proceed with the standard ESPRIT invariance equation. We form $\hat{U}_{l,\uparrow}$ (first $N-1$ rows) and $\hat{U}_{l,\downarrow}$ (last $N-1$ rows) and solve:

$$\hat{U}_{l,\uparrow} \Psi_l \approx \hat{U}_{l,\downarrow}$$

Because all $\hat{U}_l$ blocks were derived from the same U_{glob} , the resulting matrices Ψ_l are guaranteed to share the same set of eigenvectors (the columns of the global mixing matrix inverse). This satisfies the theoretical requirement for step 4 (Joint Diagonalization), allowing the algorithm to correctly pair and recover the d -dimensional frequencies.

5 Results

5.1 Phase Unwrapping Impact

In the theoretical derivation of ESPRIT-based methods, phase unwrapping is often cited as a necessary step to resolve the ambiguity of the frequency estimates when the phase arguments exceed the $(-\pi, \pi]$ range. However, we found that applying phase unwrapping in the JESPRIT context actually hurts performance.

Figure 3 compares the parameter estimation error with (left) and without (right) phase unwrapping enabled. For this experiment, the mixing matrix A was fixed to a 3×3 matrix. It can be observed that the unwrapped version consistently yields higher error rates and is robust over a smaller range of grid scale values (Δ).

5.2 Parameter Sensitivity Analysis

We evaluate the sensitivity of the JESPRIT algorithm (without phase unwrapping) to its key hyperparameters: the number of directions M , the number of snapshots S , the number of sample points per line N , and the grid scale Δ .

As shown in the subplots of Figure 3 (particularly the "Without Phase Unwrapping" panel):

- **Directions (M) and Snapshots (S):** The estimation error remains stable and low as M and S increase beyond the sufficient lower bounds (related to d and r). This suggests that the algorithm is robust to over-sampling in these dimensions, and performance does not degrade with larger values, mainly for M .
- **Samples per Line (N):** Unlike M and S , increasing N excessively can lead to performance degradation. While a certain minimum number of points is required for accuracy, very large N effectively extends the sampling range into regions where the phase arguments may exceed the principal range, causing wrapping issues when Δ is fixed.
- **Grid Scale (Δ):** This parameter exhibits a distinct "sweet spot." As discussed previously, the error is minimized when $\Delta \approx 1/\max(A)$. Deviating significantly from this value increases the estimation error.

These results highlight that while M and S can be chosen generously, N and particularly Δ require careful tuning to match the signal characteristics.

5.3 Sample Complexity and Rate Range

Finally, we analyze the sample complexity of JESPRIT—specifically, how many samples are required to successfully recover the latent factors as the problem dimensions grow. We also investigate the impact of the dynamic range of the rates in A .

To quantify this, we performed a grid search over varying ambient dimensions $d \in [1, 10]$ and ranks $r \in [1, 10]$. For each (d, r) pair, we conducted 7 independent random trials. A trial was considered successful if the average Mean Relative Error (MRE) for both rates and weights was less than or equal to 10%. We tested sample sizes of $N_s \in \{1k, 10k, 50k, 100k\}$.

Simulations were conducted only while the success rate remained above 70%. If the success rate dropped below this threshold, further simulations for higher ranks r were halted, as failure was deemed certain. This explains the absence of data points for higher r values in the heatmaps.

We compared the sample complexity for two scenarios:

1. **Small Range:** Rates A randomly drawn from $[0, 100]$.
2. **Large Range:** Rates A randomly drawn from $[0, 10000]$.

Figures 4 and 5 show the success rate (percentage of trials that met the 10% error threshold) for each configuration. The results indicate that a larger range of values in A improves the recoverability of the latent factors. With a larger dynamic range, the "directions" in the count space are more distinct, essentially providing a higher effective signal-to-noise ratio for the subspace estimation. This allows the algorithm to correctly discover more latent factors (higher r) for a given sample size compared to the small range scenario.

Furthermore, the results suggest that the sample complexity depends primarily on the number of latent factors r , rather than the ambient dimension d . As observed in the heatmaps, increasing d while keeping r constant results in a negligible increase in the sample size required for successful recovery.

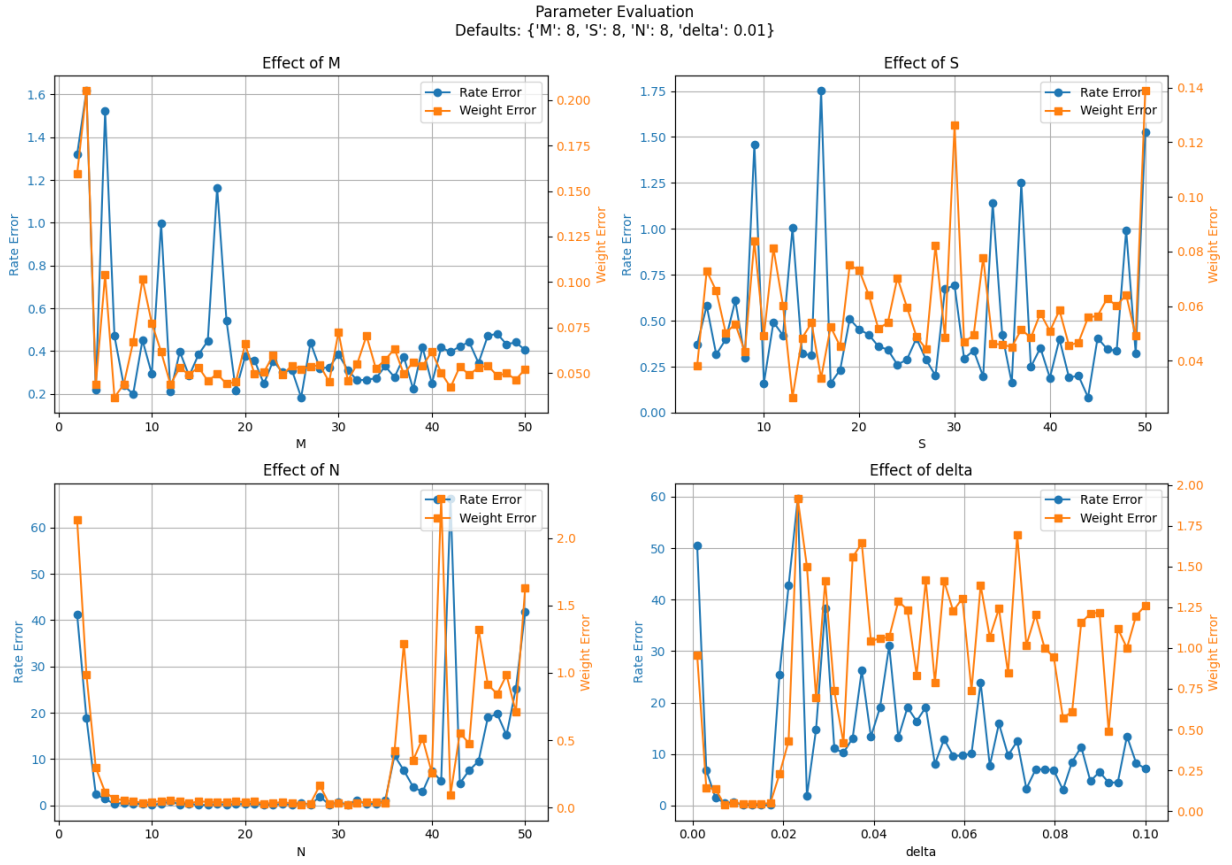


Figure 1: *
With Phase Unwrapping

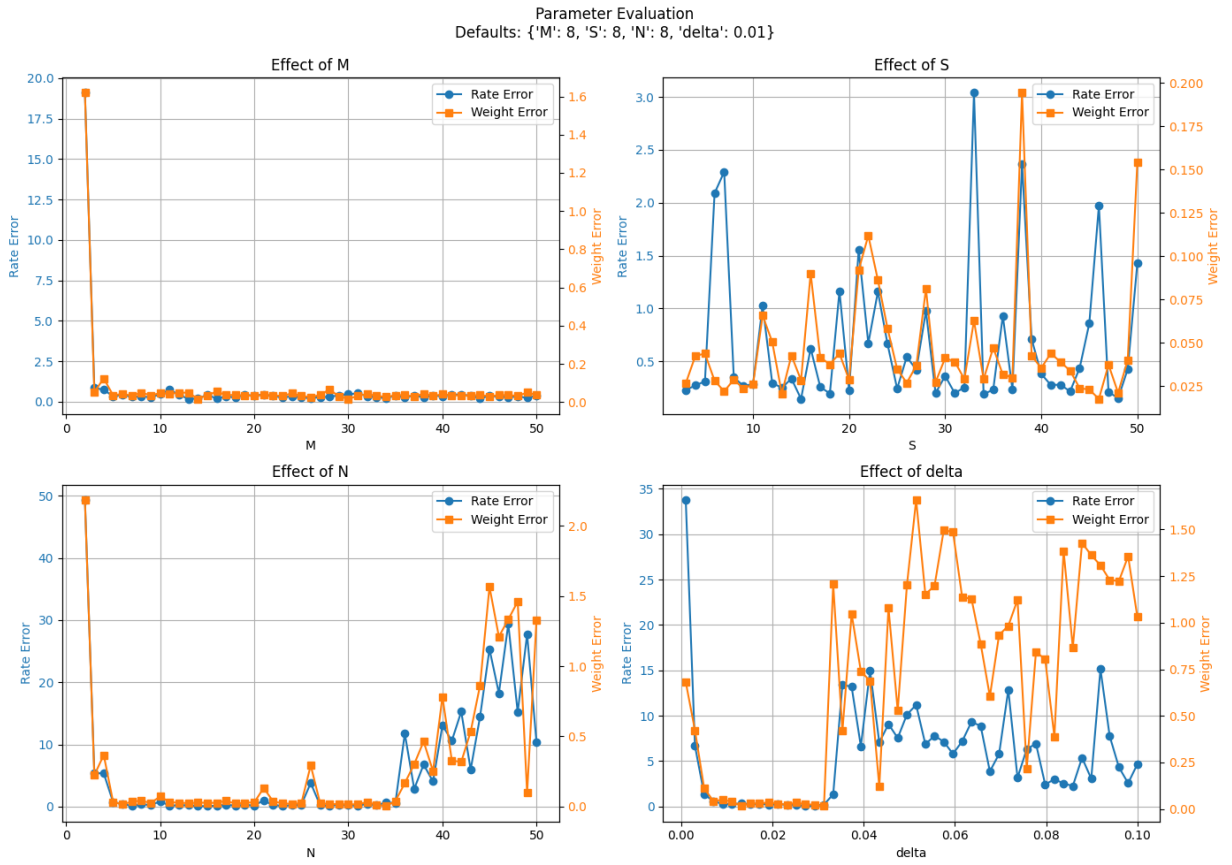


Figure 2: *
Without Phase Unwrapping

Figure 3: Comparison of estimation error with and without phase unwrapping. The unwrapped version (bottom) shows sensitivity to Δ but robustness to M and S .

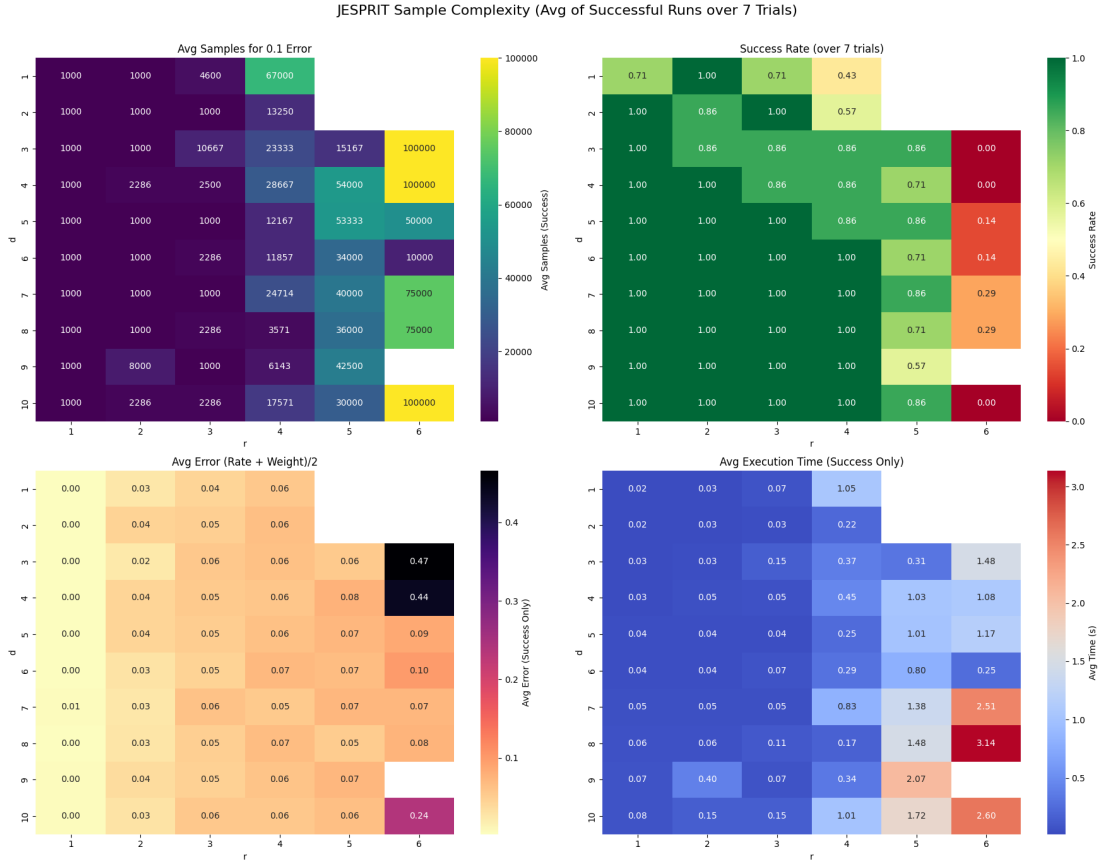


Figure 4: Sample Complexity for $A \in [0, 100]$.

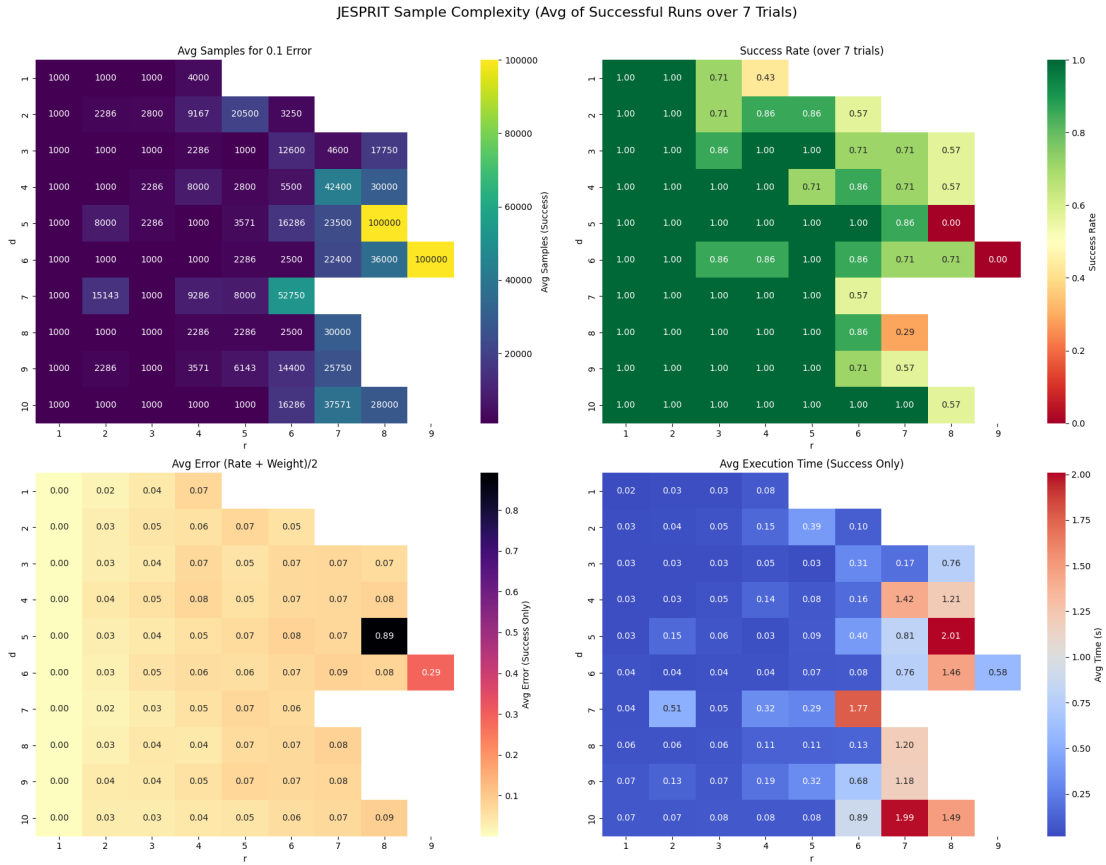


Figure 5: Sample Complexity for $A \in [0, 10000]$.

Figure 6: Comparison of Sample Complexity for different rate ranges. The larger range (bottom) allows for successful recovery of higher ranks.